

 ****Product Specification Document (PSD)****

****Audio Classification System with SE-ResNet-34 & Baseline ResNet-34****

****Project Type****: Research & Development | Machine Learning | Audio Classification

1. ****Overview****

This system implements two deep convolutional neural network architectures — ****Baseline ResNet-34**** and ****Squeeze-and-Excitation ResNet-34 (SE-ResNet-34)**** — for classifying environmental audio signals (e.g., ESC-50 dataset). The models are integrated into a Streamlit-based GUI for real-time inference, visualization, and experiment tracking via Weights & Biases (wandb). Both models are built from scratch using PyTorch and are optimized for 1D audio spectrogram inputs.

> ****Goal****: Achieve state-of-the-art performance on environmental sound classification while evaluating the impact of Squeeze-and-Excitation (SE) attention mechanisms on model accuracy and convergence.

2. ****System Architecture****

2.1 Model Components

Component	Description
----- -----	
Input	Log-Mel Spectrograms (1-channel, 128×128 or configurable) from audio files (e.g., ESC-50)
Feature Extractor	ResNet-34 backbone with 34 layers (convolutional blocks)
Attention Module	SEBlock (Squeeze-and-Excitation) — applied after each residual block in SE-ResNet-34
Output	50-class softmax layer (for ESC-50); configurable for other datasets
Normalization	BatchNorm2d after each convolutional layer
Activation	ReLU

2.2 Model Variants

Variant	Architecture	Block Configuration	SE Attention	Parameters (approx.)
----- ----- ----- ----- -----				
Baseline ResNet-34	Standard ResNet-34	[3, 4, 6, 3]	✗ No	~21.3M
SE-ResNet-34	ResNet-34 + SE Blocks	[3, 4, 6, 3]	✓ Yes (per block)	~21.5M

> ✓ **Note**: SE-ResNet-34 adds negligible parameters (~0.1M) due to lightweight fully connected layers in SEBlock.

2.3 SEBlock Implementation

```
```python
```

```
class SEBlock(nn.Module):
```

```
 def __init__(self, channel, reduction_ratio=16):
```

```
 super(SEBlock, self).__init__()
```

```
 self.avg_pool = nn.AdaptiveAvgPool2d(1)
```

```
 self.fc = nn.Sequential(
```

```
 nn.Linear(channel, channel // reduction_ratio, bias=False),
```

```
 nn.ReLU(inplace=True),
```

```
 nn.Linear(channel // reduction_ratio, channel, bias=False),
```

```
 nn.Sigmoid()
```

```
)
```

```
 def forward(self, x):
```

```
 b, c, _, _ = x.size()
```

```
 y = self.avg_pool(x).view(b, c)
```

```
 y = self.fc(y).view(b, c, 1, 1)
```

```
 return x * y.expand_as(x)
```

```
```
```

- ****Reduction Ratio****: Configurable (default: 16)

- ****Function****: Dynamically recalibrates channel-wise feature responses based on global context.

```
---
```

3. **Implementation Details**

3.1 Core Models (Code Structure)

- **models/se_resnet.py**:
 - Defines `SEBasicBlock`, `SEBlock`, and `SEResNet` class
 - Exports two functions:
 - `se_resnet18()` → *Deprecated (for reference only)*
 - `se_resnet34()` →  **Primary variant used**
- **models/baseline_model.py**:
 - Contains `BasicBlock` and `baseline_resnet34()`
 - No attention modules; standard residual connections only

3.2 Model Instantiation (config-driven)

```
```python
In training/inference script

if config.model.name == "baseline_resnet":

 model = baseline_resnet34(num_classes=config.model.num_classes)

elif config.model.name == "se_resnet":

 model = se_resnet34(num_classes=config.model.num_classes,

 reduction_ratio=config.model.get('reduction_ratio', 16))
```

else:

```
 raise ValueError(f"Unknown model name: {config.model.name}")
```

```
'''
```

>  **\*\*All models now use ResNet-34 architecture\*\*** — no ResNet-18 used in production.

```

```

#### ## 4. **\*\*Training & Evaluation\*\***

Parameter	Value
-----------	-------

-----	-----
-------	-------

<b>**Dataset**</b>	ESC-50 (40 seconds, 50 classes, 2000 samples)
--------------------	-----------------------------------------------

<b>**Input Shape**</b>	`(1, 128, 128)` — Log-Mel Spectrogram
------------------------	---------------------------------------

<b>**Optimizer**</b>	AdamW (lr=1e-3, weight_decay=1e-4)
----------------------	------------------------------------

<b>**Scheduler**</b>	StepLR (step_size=10, gamma=0.5)
----------------------	----------------------------------

<b>**Batch Size**</b>	32
-----------------------	----

<b>**Epochs**</b>	50 (early stopping if no improvement for 10 epochs)
-------------------	-----------------------------------------------------

<b>**Loss**</b>	CrossEntropyLoss
-----------------	------------------

<b>**Metrics**</b>	Accuracy, F1-Score, Confusion Matrix, Precision/Recall
--------------------	--------------------------------------------------------

<b>**Hardware**</b>	NVIDIA GPU (e.g., RTX 3060 or higher)
---------------------	---------------------------------------

```

```

## ## 5. **Integration & Deployment**

Component	Description
-----------	-------------

--	--

<b>GUI</b>	Streamlit web interface for: - Uploading audio files - Real-time prediction - Spectrogram visualization - Model selection (baseline vs SE)
------------	--------------------------------------------------------------------------------------------------------------------------------------------------------

<b>Experiment Tracking</b>	Weights & Biases (wandb) for: - Loss/accuracy curves - Hyperparameter logging - Model versioning - Gradient & weight histograms
----------------------------	---------------------------------------------------------------------------------------------------------------------------------------------

<b>Model Export</b>	.pt and .onnx formats supported for deployment
---------------------	------------------------------------------------

<b>Inference Speed</b>	~15ms per sample on GPU
------------------------	-------------------------

---

## ## 6. **Configuration File Example (config.yaml)**

```
```yaml
```

```
model:
```

```
  name: "se_resnet"      # "baseline_resnet" or "se_resnet"
```

```
  num_classes: 50
```

```
  reduction_ratio: 16    # Only used for se_resnet
```

```
training:
```

batch_size: 32

epochs: 50

lr: 0.001

scheduler_step: 10

scheduler_gamma: 0.5

data:

dataset: "esc50"

sr: 22050

n_mels: 128

hop_length: 512

max_duration: 5.0

wandb:

project: "audio-se-resnet-34"

entity: "shivansh-bansal-piet"

log_gradients: true

``

7. **Expected Outcomes & KPIs**

Metric	Target
Baseline ResNet-34 Accuracy	$\geq 82\%$
SE-ResNet-34 Accuracy	$\geq 85\%$ (target improvement: +3–5%)
Training Convergence	Faster convergence with SE due to feature recalibration
Model Robustness	Improved performance on noisy/low-SNR samples
Ablation Study	Quantify SEBlock contribution via ablation (with/without SE)

8. Future Enhancements

- ☐ Add data augmentation: SpecAugment, time/frequency masking
- ☐ Support multi-class, multi-label classification
- ☐ Integrate Grad-CAM for attention visualization
- ☐ Deploy as REST API (FastAPI)
- ☐ Optimize for edge devices (TensorRT, ONNX Runtime)

9. Dependencies

```
```yaml
```

torch>=2.0

torchvision>=0.15

numpy>=1.21

librosa>=0.10

scikit-learn>=1.3

streamlit>=1.30

wandb>=0.16

``

---

## 10. **Author & Contact**

- **Name**: Shivansh Bansal

- **Affiliation**: Panipat Institute of Engineering and Technology

- **Email**: [your.email@piet.edu.in]

- **GitHub**: [github.com/shivanshbansal] *\*(optional)\**

- **Project Status**: **Active Research**